

EXPERIMENT NO :- 01

AIM:- To study various wireless technologies, their working principles, frequency ranges, and applications in modern communication.

1. Bluetooth

Bluetooth offers communication ranges from 10 to 100 meters, depending on the version (Bluetooth 4.0 and Bluetooth 5). It is commonly used for short-range connections in personal area networks (PANs) like smartphones, headphones, and wearables. Bluetooth modules are cost-effective, typically priced between \$1 to \$10, making them affordable for consumer and IoT devices. Bluetooth 4.0 is widely used in low-cost products, while Bluetooth 5 offers greater range and speed for newer devices. Bluetooth Low Energy (BLE) technology optimizes power consumption, extending the battery life of devices like smartwatches and fitness trackers. Throughput for Bluetooth Classic ranges from 1 Mbps to 3 Mbps, and Bluetooth 5 can reach up to 50 Mbps, supporting applications such as audio streaming and file transfer. Bluetooth typically allows for a fan-out of 7 devices in a piconet but, with Bluetooth Mesh, this can support a much larger number of devices, making it ideal for smart home applications.

2. Wi-Fi

Wi-Fi supports communication ranges from 30 meters to 100 meters in open spaces, with performance influenced by obstacles and interference. Wi-Fi modules cost between \$10 and \$50, depending on the capabilities of the version (e.g., Wi-Fi 5 vs. Wi-Fi 6). Wi-Fi 5 and Wi-Fi 6 are common in modern routers and IoT devices, with Wi-Fi 6 offering faster speeds and better efficiency for dense environments. Wi-Fi is suitable for home and office networks and applications like internet browsing, HD/4K video streaming, and online gaming. Wi-Fi has a medium propagation delay of 5ms to 30ms, and its throughput ranges from 50 Mbps to 10 Gbps in Wi-Fi 6, making it ideal for high-speed internet and video calls. While Wi-Fi consumes more power than Bluetooth or Zigbee, it supports high fan-out, capable of connecting hundreds of devices to a single access point, making it ideal for enterprise networks.

3. Zigbee

Zigbee is typically used in home automation and smart lighting, with a communication range of 10 to 100 meters. The cost of Zigbee modules is low, ranging from \$2 to \$10, which makes it ideal for affordable IoT devices. Zigbee's low power consumption is specifically optimized for battery-operated devices such as smart home sensors and remote controls. It has a low propagation delay, typically between

10ms to 30ms, which makes it suitable for low-latency automation tasks. Zigbee supports low throughput (20 Kbps to 250 Kbps) and is commonly used for sending small amounts of data like sensor readings or status updates. Zigbee is highly scalable and supports a high fan-out through its mesh network, which can connect a large number of devices, ideal for home automation systems.

4. LoRa (Long Range)

LoRa provides long-range communication, typically up to 15 km in open space, making it well-suited for remote IoT applications like agricultural sensors and environmental monitoring. LoRa devices cost between \$10 and \$50, which is reasonable for many IoT projects, though more specialized hardware may be more expensive. LoRa operates with a moderate propagation delay of around 100ms and consumes very little power, making it ideal for battery-powered devices deployed in remote locations. The throughput of LoRa is limited (0.3 kbps to 27 kbps), making it suitable for small packet transmissions such as environmental data or device status updates. LoRa supports moderate fan-out, typically used in regional IoT networks where devices connect to a gateway.

5. 5G

5G networks offer high-speed internet and ultra-low-latency applications with communication ranges from 1 km to 5 km. Devices equipped with 5G modules range in cost from \$100 to \$500, depending on the manufacturer and use case. 5G offers a very low propagation delay (around 1ms), making it ideal for applications like autonomous vehicles, real-time remote monitoring, and virtual reality. Due to the high speeds and low latency, power consumption is higher than with other technologies like Bluetooth or Zigbee, making it less suited for battery-powered devices. 5G offers throughput ranging from 1 Gbps to 10 Gbps, supporting applications such as 4K/8K video streaming, augmented reality (AR), and virtual reality (VR). With its ability to support millions of devices in dense environments, 5G is well-suited for smart cities and large-scale IoT deployments.

6. NFC (Near Field Communication)

NFC is designed for very short-range communication, typically within 10 cm. The cost of NFC devices is very low, typically ranging from \$1 to \$5, making them affordable for applications like contactless payments and device pairing. With a very low propagation delay (under 1ms), NFC is ideal for instant data transfer, allowing for quick exchanges between devices. NFC operates with low power consumption, especially when using passive tags, which is common in inventory management and

similar applications. The throughput is relatively low, ranging from 106 kbps to 424 kbps, making NFC ideal for applications that require quick, low-volume data exchange. The fan-out is extremely limited, as NFC allows only one device to communicate with another at a time, suitable for one-to-one interactions like payments or device pairing.

7. Infrared (IR)

Infrared (IR) communication typically operates over a range of 1 to 5 meters and is commonly used for remote controls and simple data transfer. IR devices are inexpensive, costing between \$1 and \$10, and are used in applications like TV remote controls, sensor communication, and simple data exchange between devices. The propagation delay for IR is typically low, generally under 5ms. While IR consumes low to medium power, depending on the distance and settings, it is not suitable for high-speed data transfer. Throughput can reach up to 4 Mbps with higher-end IR devices, such as those using the IRDA protocol. IR communication is typically point-to-point and supports low fan-out, meaning it is used for one-to-one communication between devices.

8. Satellite Communication

Satellite communication offers global coverage, with communication ranges spanning thousands of kilometers. It is used for remote communication, internet access in rural areas, and emergency services. Devices for satellite communication are generally expensive, typically ranging from several hundred dollars to several thousand dollars, depending on the technology and purpose (e.g., satellite phones or satellite internet terminals). The propagation delay for satellite communication is relatively high (500ms to 1000ms), due to the long distances the signals must travel to and from satellites in geostationary or low Earth orbit. Throughput for satellite communication can range from 1 Mbps to several Gbps, depending on the technology used, and satellite communication is capable of supporting very high fan-out, enabling communication with a wide range of users across the globe.

9. Cellular Communication (2G, 3G, 4G, LTE, 5G)

Cellular networks typically offer communication ranges from 1 km to 10 km, depending on the cellular standard (e.g., 2G, 3G, 4G, 5G) and the placement of cell towers. The cost of cellular communication modules ranges from \$20 to \$200 or more, depending on the technology (e.g., LTE or 5G). Cellular communication provides voice services, internet browsing, video streaming, and gaming, with a propagation delay ranging from 50ms to 150ms for 3G and 4G, and down to 1ms for

5G. Power consumption varies by technology, with 5G generally requiring more power than earlier standards like 2G or 3G. Throughput can range from 100 kbps in 2G to 1 Gbps in 5G, supporting mobile data services and IoT applications. Cellular networks can support very high fan-out, with modern cellular networks able to handle millions of devices within a given cell, making them ideal for large-scale mobile and IoT applications.

PARAMETERS

1.Range of Communication:

This refers to the maximum distance over which a device or technology can communicate effectively. For wireless technologies, the range depends on factors like signal strength, environmental conditions, and interference. For example, Wi-Fi might have a range of a few hundred feet indoors, while cellular networks can cover miles.

2.Cost of the Devices for that Technology:

The cost refers to the price of the devices required for communication using a specific technology. This could include the cost of transmitters, receivers, routers, or specialized devices (e.g., 5G base stations). Higher technology sophistication usually correlates with higher device costs.

3.Propagation Delay:

Propagation delay is the time it takes for a signal to travel from the sender to the receiver. This delay is influenced by the distance between the devices and the medium through which the signal travels. For instance, fiber-optic networks have lower propagation delays compared to satellite communication due to the distance signals must travel.

4.Power Consumption:

Power consumption refers to the amount of electrical energy a device or technology uses while in operation. Efficient power use is crucial for mobile devices and large-scale networks, especially for IoT devices, which often rely on battery power.

5.Throughput:

Throughput is the rate at which data is successfully transmitted from one point to another over a network or communication system. It is usually measured in bits per second (bps) or multiples like Mbps or Gbps. Higher throughput is essential for applications that require large amounts of data to be transferred quickly, such as video streaming or cloud services.

6.Fan Out: Fan out refers to the number of devices or users a single device or communication node can serve or connect to. In networking, a higher fan-out indicates that a node or device can handle a larger number of simultaneous connections.

Wireless Communication Technologies Comparison Table

Technology	Range of Communication	Cost of Device	Propagation Delay	Power Consumption	Throughput	Fan Out
Bluetooth	10m to 100m (depending on version)	\$1 to \$10	1-5ms	Low	1-3 Mbps (Bluetooth Classic) / 50 Mbps (Bluetooth 5)	Limited (typically 7 devices in piconet)
Wi-Fi	30m to 100m (depending on environment)	\$10 to \$50	5-30ms	Medium	50 Mbps to 10 Gbps (Wi-Fi 6)	High (many devices connected to access point)
Zigbee	10m to 100m	\$2 to \$10	10-30ms	Very Low	20 Kbps to 250 Kbps	High (supports many devices in mesh)
LoRa	Up to 15 km in open space	\$10 to \$50	~100ms	Very Low	0.3 kbps to 27 kbps	Moderate (depends on network infrastructure)
5G	1-5 km	\$100 to \$500	1ms	High	1 Gbps to 10 Gbps	Very High (supports millions of devices)
NFC	Typically 10 cm or less	\$1 to \$5	<1ms	Very Low	106 kbps to 424 kbps	Low (point-to-point)
Infrared (IR)	Typically 1m to 5m	\$1 to \$10	<5ms	Low to Medium	Up to 4 Mbps	Low (point-to-point)
Satellite Communication	Global (thousands of kilometers)	Several hundred dollars and more	500ms to 1000ms	High	1 Mbps to several Gbps	Very High (supports global coverage)
Cellular (2G, 3G, 4G, LTE, 5G)	1-10 km	\$20 to \$200	50ms to 150ms (2G/3G/4G), 1ms (5G)	Medium to High	100 kbps (2G) to 1 Gbps (5G)	Very High (supports large networks)

Conclusion: -

Wireless technologies are essential in modern communication, facilitating seamless data transfer without the need for physical connections. These technologies encompass a wide range of applications, each designed to serve specific purposes based on factors such as range, frequency, power consumption, and intended use.

For short-range communication, technologies like Near Field Communication (NFC), Bluetooth, and ZigBee are commonly used. NFC enables contactless interactions over very short distances, making it ideal for secure transactions and access control. Bluetooth is widely implemented in personal devices, such as wireless headphones, smartwatches, and fitness trackers, offering moderate-range connectivity with low power consumption. ZigBee, on the other hand, is particularly useful in home automation and industrial applications due to its ability to support mesh networking and low-power operation.

For long-range communication, technologies like LoRa, satellite communication, and 5G play a crucial role. LoRa (Long Range) is designed for low-power, wide-area networks (LPWANs) and is commonly used in IoT applications, such as smart agriculture and environmental monitoring. Satellite communication enables global coverage, making it indispensable for remote locations, maritime communication, and disaster recovery scenarios. Meanwhile, 5G technology provides ultra-fast data transfer speeds, low latency, and high capacity, revolutionizing industries such as autonomous vehicles, smart cities, and telemedicine.

Understanding the characteristics and capabilities of these wireless technologies is vital in choosing the most suitable option for specific applications, ensuring efficiency, reliability, and optimal performance in various communication scenarios.